



LESSON 4-5

Solving Rational Equations

Lesson Overview

Objective

Students will be able to:

- ✓ Solve rational equations in one variable.
- ✓ Identify extraneous solutions to rational equations and give examples of how they arise.

Essential Understanding

Rational equations contain a rational expression and can be solved by multiplying each side of the equation by a common denominator to eliminate the fractions. Any solution that is excluded from the domain of the original equation is extraneous.

Previously in this topic, students:

- Added, subtracted, multiplied, and divided rational expressions.

In this lesson, students:

- Solve rational equations by multiplying each side of the equation by a common denominator and identifying extraneous solutions.

In later topics, students will:

- Add, subtract, multiply, and divide rational expressions as part of the process for solving exponential and logarithmic equations.

This lesson emphasizes a blend of *procedural skill and fluency* and *application*.

- Students solve rational equations by multiplying each side of the equation by a common denominator to eliminate fractions.
- Students apply their knowledge of operations with rational expressions to solve rate problems.



Glossary is available at SavvasRealize.com.

Vocabulary Builder

REVIEW VOCABULARY

English | Spanish

- **domain** | *dominio*
- **rational expression** | *expresión racional*
- **real numbers** | *número real*
- **undefined** | *undefined*

NEW VOCABULARY

- **extraneous solution** | *solución extraña*
- **rational equation** | *ecuación racional*

VOCABULARY ACTIVITY

Review the term *undefined* as it relates to division by zero. Introduce the new term *extraneous solution* by asking students to consider the word *extraneous* by thinking about what extra means.

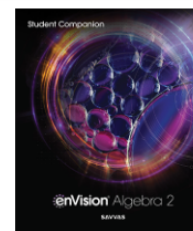
Q: What is an example of extraneous information?

Q: Think about what you brought with you to math class today. What materials do you have with you that are extraneous?

Explain that an *extraneous solution* is a solution that is not valid because it is excluded from the *domain* of the original equation. Solutions that cause terms of the original equation to be undefined are extraneous solutions.

Student Companion

Students can do their work for the lesson in their *Student Companion* or on Savvas Realize.



Mathematics Overview

Students find solutions to rational equations in one variable. They multiply both sides of the equation by a common denominator and then solve for the variable. Students also find extraneous solutions by identifying values that are excluded from the domain of the original equation.

Applying Math Practices

Make Sense of Problems and Persevere in Solving Them

Students recognize that they can solve rational equations by multiplying both sides of an equation by the same factors, as long as none of the factors are zero.

Look for and Make Use of Structure

Students connect mathematical ideas and concepts when they recognize that they can use asymptotes to identify extraneous solutions.

STEP 1 Explore



CRITIQUE & EXPLAIN

INSTRUCTIONAL FOCUS Students will analyze and critique two different methods of solving an equation with fractions. These methods will be helpful when students solve rational equations.

STUDENT COMPANION Students can complete the *Critique & Explain* activity in their *Student Companion*.

Before **WHOLE CLASS**

Implement Tasks that Promote Reasoning and Problem Solving **ETP**

Q: What do you notice about the equation? [The equation is a linear equation. The coefficient of the x-term and the constants are fractions.]

During **SMALL GROUP**

Support Productive Struggle in Learning Mathematics **ETP**

Q: What strategy does Nicky use to solve the equation? [Nicky uses a similar strategy as she would use to solve a linear equation with integers. She subtracts to isolate the variable term on one side of the equation; then she divides to isolate the variable.]

Q: What strategy does Tavon use to solve the equation? [Tavon multiplies each side of the equation by 10. Because each side of the equation is multiplied by the same value, the result is an equivalent equation.]

For Early Finishers

Q: Solve the equation $\frac{2}{3}x - \frac{4}{5} = 1\frac{1}{5}$ using the methods that Nicky and Tavon use. Does each method result in the same solution? [The solution is $x = 3$. Each method of solving this problem results in the same answer.]

After **WHOLE CLASS**

Facilitate Meaningful Mathematical Discourse **ETP**

Q: What do you notice if you rewrite each fraction with a denominator of 10? [The numbers in the numerators are the same as those in the equation that Tavon found after multiplying both sides of the equation by 10.]

Q: Substitute $x = 1$ into $\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}$ and $5x + 4 = 9$. What does the result tell you about the equations? [When you substitute $x = 1$ into both equations, the equations are true. These are equivalent equations.]

Use with CRITIQUE & EXPLAIN

HABITS OF MIND

Reason If Tavon multiplied each side of the equation by 100, would his answer have been 10 times as much? Explain.

[No; multiplying each side of an equation by the same value does not change the solution of the equation.]

Critique & Explain is available at SavvasRealize.com.

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CRITIQUE & EXPLAIN

46 Nicky and Tavon used different methods to solve the equation $\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}$.

Nicky	Tavon
$\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}$	$\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}$
$\frac{1}{2}x = \frac{9}{10} - \frac{2}{5}$	$10(\frac{1}{2}x + \frac{2}{5} = \frac{9}{10})$
$\frac{1}{2}x = \frac{9}{10} - \frac{4}{10}$	$5x + 4 = 9$
$\frac{1}{2}x = \frac{5}{10}$	$5x = 5$
$x = 1$	$x = 1$
The solution is 1.	The solution is 1.

A. Explain the different strategies that Nicky and Tavon used, and the advantages or disadvantages of each.

Enter your answer:

47 B. Did Nicky use a correct method to solve the equation? Did Tavon?

Enter your answer:

48 C. **Use Structure** Why might Tavon have chosen to multiply both sides of the equation by 10? Could he have used another number? Explain.

STUDENT EDITION

4-5
Solving Rational Equations

CRITIQUE & EXPLAIN

Nicky and Tavon used different methods to solve the equation $\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}$.

Nicky	Tavon
$\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}$	$\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}$
$\frac{1}{2}x = \frac{9}{10} - \frac{2}{5}$	$10(\frac{1}{2}x + \frac{2}{5} = \frac{9}{10})$
$\frac{1}{2}x = \frac{9}{10} - \frac{4}{10}$	$5x + 4 = 9$
$\frac{1}{2}x = \frac{5}{10}$	$5x = 5$
$x = 1$	$x = 1$
The solution is 1.	The solution is 1.

A. Explain the different strategies that Nicky and Tavon used and the advantages or disadvantages of each.

B. Did Nicky use a correct method to solve the equation? Did Tavon?

C. **Use Structure** Why might Tavon have chosen to multiply both sides of the equation by 10? Could he have used another number? Explain.

SAMPLE STUDENT WORK

- A. Tavon multiplied the whole equation by 10, which gets rid of the fractions but also adds a step to solving the problem. Nicky kept the fractions, which meant having to find common denominators to do some of the solving.
- B. Both methods are correct.
- C. Tavon used 10 because it is the LCD of the fractions. He could have used another multiplier that is a common multiple of the denominators and still gotten a correct answer, but it would mean more reducing.



INTRODUCE THE ESSENTIAL QUESTION

Establish Mathematics Goals to Focus Learning **ETP**

Introduce students to rational equations. Remind them that they have added, subtracted, multiplied, and divided rational expressions. This helps students solve rational equations and identify extraneous solutions. Explain that they can use rational equations to solve rate problems.

EXAMPLE 1 Solve a Rational Equation

Pose Purposeful Questions **ETP**

Q: Why do you have to confirm that the solution is valid in the original equation? [You have to make sure the solution does not cause the denominator of the rational expression to equal zero.]

Try It! Answers

- $x = -\frac{9}{2}$
- $x = \frac{15}{2}$

Elicit and Use Evidence of Student Thinking **ETP**

Q: Why do you multiply both sides of the equation by the common denominator? [You multiply both sides by the common denominator to eliminate the fractions.]

Examples are available at SavvasRealize.com.

ESSENTIAL QUESTION

How can you solve rational equations and identify extraneous solutions?

EXAMPLE 1

Solve a Rational Equation

What is the solution to each rational equation?

A. $\frac{1}{x+4} = 2$

$(x+4)\left(\frac{1}{x+4}\right) = 2(x+4)$

$1 = 2x + 8$

$x = -\frac{7}{2}$

The solution is $x = -\frac{7}{2}$.

B. $\frac{1}{x-3} = 5$

$(x-3)\left(\frac{1}{x-3}\right) = 5(x-3)$

$1 = 5x - 15$

$x = \frac{16}{5}$

The solution is $x = \frac{16}{5}$.

VOCABULARY

A **rational equation** is an equation that contains a rational expression.

Multiply both sides of the equation by the common denominator to eliminate the fractions. Then solve. Confirm that the solution is valid in the original equation.

Try It! 1. What is the solution to each equation?

a. $\frac{2}{x+5} = 4$ b. $\frac{1}{x-7} = 2$

HABITS OF MIND

Use with EXAMPLE 1

Critique Reasoning For part (b), Kaitlyn wrote $1 = 2x - 7$, then $8 = 2x$, and $4 = x$. Is she correct? Explain.

[No; Kaitlyn did not distribute the 2 properly. The equation becomes $1 = 2(x - 7)$, with $x = 7.5$.]

Additional Examples

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ADDITIONAL EXAMPLE 3

ESSENTIAL QUESTION

What are the solution(s) to the equation?

$\frac{5}{x+6} + \frac{4}{x+3} = 3$

$\frac{5}{x+6} + \frac{4}{x+3} = 3$

$(x+6)(x+3)\left(\frac{5}{x+6} + \frac{4}{x+3}\right) = 3(x+6)(x+3)$

$5(x+3) + 4(x+6) = 3(x+6)(x+3)$

$5x + 15 + 4x + 24 = 3x^2 + 27x + 54$

$0 = x^2 + 6x + 5$

$0 = (x+1)(x+5)$

$x = -1$ and $x = -5$

Both -1 and -5 are in the domain of the original equation, so the solutions are $x = -1$ and $x = -5$.

SOLUTION

Example 3 Help students understand when a solution is extraneous.

Q: Does a rational equation with multiple solutions always have an extraneous solution? [No, it is possible that each solution is in the domain of the original equation.]

Additional Examples are available at SavvasRealize.com.

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ADDITIONAL EXAMPLE 5

ESSENTIAL QUESTION

It takes Reggie 60 minutes to wash a car by himself. It takes Angie 40 minutes to wash a car by herself. It takes Eugene 30 minutes to wash a car by himself. How long does it take them to wash the car together?

Reggie's rate is $\frac{1}{60}$ car per minute, and his rate is $\frac{1}{60}$ car per minute. Eugene's rate is $\frac{1}{30}$ car per minute, and his rate is $\frac{1}{30}$ car per minute. Angie's rate is $\frac{1}{40}$ car per minute, and her rate is $\frac{1}{40}$ car per minute.

Together, they wash 1 car in 30 minutes, so the sum of their rates is equal to $\frac{1}{30}$.

$\frac{1}{60} + \frac{1}{40} + \frac{1}{30} = \frac{1}{30}$

$\frac{2}{120} + \frac{3}{120} + \frac{4}{120} = \frac{1}{30}$

$\frac{9}{120} = \frac{1}{30}$

$\frac{3}{40} = \frac{1}{30}$

$30 \cdot \frac{3}{40} = 30 \cdot \frac{1}{30}$

$\frac{9}{4} = 1$

$9 = 4$

$5 = 4 - 9$

$5 = -5$

$x = -5$ is not a solution because it is not in the domain of the original equation.

Reggie and Eugene wash the car in 30 minutes.

SOLUTION

Example 5 Extend students' understanding by relating the rates people take to complete a task to rational expressions.

Q: What do the rational expressions in the original equation represent? [$\frac{1}{40}$, $\frac{1}{x-5}$, and $\frac{1}{x}$ represent individual rates to wash a car, and $\frac{1}{20}$ represents the rate for all three to wash a car together.]

STEP 2 Understand & Apply



EXAMPLE 2 Solve a Work-Rate Problem

Pose Purposeful Questions ETP

Q: Why do you add the expressions that represent the two work-rates?

[In one hour, Arthur completes $\frac{1}{x}$ of the job, and Cheyenne completes $\frac{2}{x}$ of the job. Together, they complete $\frac{1}{x} + \frac{2}{x}$ of the job.]

Q: How could you solve this equation without multiplying both sides by $24x$?

[You can add the fractions and then cross-multiply.]

Try It! Answer

2. 48 hours

Elicit and Use Evidence of Student Thinking ETP

Q: Why is it important to define the variable before writing an equation to represent the problem?

[Without understanding what the variable represents, it would be difficult to set up the equation correctly.]

Use with EXAMPLE 2

HABITS OF MIND

Reason In a work-rate problem, explain why you can't average the individual rates to determine how long it will take to complete a job together.

[Averaging the two rates would result in a rate less than the faster person's rate. But two individuals working together would complete the work in less time than the person with the faster rate could do alone.]

Examples are available at SavvasRealize.com.

APPLICATION

EXAMPLE 2 Solve a Work-Rate Problem

Arthur and Cheyenne are making Chinese paper cutting designs to give as gifts to wish family and friends good luck in the coming year. Together they can complete the job in 24 hours. How long would it take Cheyenne to complete the job if she was working alone?



Step 1 Determine the work-rates of Arthur and Cheyenne.

Let x represent the number of hours Arthur needs to finish the job himself.

Arthur works at a rate of $\frac{1 \text{ job}}{x \text{ hours}} = \frac{1}{x} \text{ jobs per hour}$.

The amount of a job completed per hour is the work-rate.

Cheyenne is twice as fast, so she works at a rate of $\frac{1 \text{ job}}{\frac{1}{2}x \text{ hours}} = \frac{2}{x} \text{ jobs per hour}$.

Working together, they complete the job in 24 hours, so the work-rate is $\frac{1 \text{ job}}{24 \text{ hours}} = \frac{1}{24} \text{ jobs per hour}$.

Step 2 Write the equation for their rates working together.

$$\frac{1}{x} + \frac{2}{x} = \frac{1}{24}$$

$$24x\left(\frac{1}{x} + \frac{2}{x}\right) = \left(\frac{1}{24}\right)24x$$

$$24 + 48 = x$$

$$72 = x$$

It takes Arthur 72 hours to create the gifts alone. Since Cheyenne works twice as fast as Arthur, it would take her 36 hours to create the gifts alone.

Try It! 2. It takes 12 hours to fill a pool with two pipes, where the water in one pipe flows three times as fast as the other pipe. How long will it take the slower pipe to fill the pool by itself?



Extend Student Thinking

USE WITH EXAMPLE 2 Challenge students to solve a work-rate problem with radical solutions.

Q: Rebecca can rake all of the leaves in a yard in x hours. Angus can rake all of the leaves in a yard in 2 more hours than Rebecca. Together they can rake the yard in 10 hours. How many hours does it take for Rebecca to rake the yard by herself? Write your answer to the nearest hundredth. [19.05 h]



EXAMPLE 3 Identify an Extraneous Solution

Use and Connect Mathematical Representations ETP

Q: How do you know that the common denominator $(x - 5)(x - 3)$ will divide out $x^2 - 8x + 15$?

[The product of $x - 5$ and $x - 3$ is equal to $x^2 - 8x + 15$.]

Q: Are all values that make the value of the denominator 0 extraneous solutions?

[No; in this equation, 3 is a value that makes the denominator 0, but it is not an extraneous solution of the equation.]

Try It! Answer

3. no solution

Elicit and Use Evidence of Student Thinking ETP

Q: How can you use a graph to verify your solution?

[Graph each side of the equation. Notice that the two graphs do not intersect. Therefore, there is no solution.]



Examples are available at SavvasRealize.com.

CONCEPTUAL UNDERSTANDING

EXAMPLE 3 Identify an Extraneous Solution

What is the solution of the equation $\frac{1}{x-5} + \frac{x}{x-3} = \frac{2}{x^2 - 8x + 15}$?

Step 1 Multiply each side of the equation by the common denominator, $(x - 5)(x - 3)$.

$$(x - 5)(x - 3) \left(\frac{1}{x-5} + \frac{x}{x-3} \right) = \frac{2(x-5)(x-3)}{x^2 - 8x + 15}$$

Step 2 Continue to simplify.

$$\frac{(x-5)(x-3)}{x-5} + \frac{x(x-5)(x-3)}{x-3} = \frac{2(x-5)(x-3)}{x^2 - 8x + 15}$$

Step 3 Divide out common factors in the numerator and the denominator.

$$\frac{(x-5)(x-3)}{x-5} + \frac{x(x-5)(x-3)}{x-3} = \frac{2(x-5)(x-3)}{x^2 - 8x + 15}$$

Step 4 Solve the equation.

$$(x - 3) + x(x - 5) = 2$$

$$x - 3 + x^2 - 5x = 2$$

$$x^2 - 4x - 3 = 2$$

$$x^2 - 4x - 5 = 0$$

$$(x - 5)(x + 1) = 0$$

$$x = 5 \text{ and } x = -1$$

You can divide out common factors under the assumption that $\frac{x-5}{x-5} = 1$ and $\frac{x-3}{x-3} = 1$. This is only true if $x \neq 5$ or 3 .

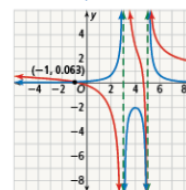
Consider both solutions. If either solution makes the value of the denominator 0, it is not valid.

The solution $x = 5$ is an **extraneous solution** because it is not in the domain of the original equation.

The solution of the equation $\frac{1}{x-5} + \frac{x}{x-3} = \frac{2}{x^2 - 8x + 15}$ is -1 .

To see that there is only one solution, graph $f(x) = \frac{1}{x-5} + \frac{x}{x-3}$ and

$$g(x) = \frac{2}{x^2 - 8x + 15}$$



Both graphs have vertical asymptotes at $x = 3$ and $x = 5$. Therefore, neither graph has a value at $x = 5$. The graphs only intersect in one point, at $x = -1$.

LOOK FOR RELATIONSHIPS
How are an extraneous solution and an asymptote related? Is this always true?

Try It! 3. What is the solution to the equation $\frac{1}{x+2} + \frac{1}{x-2} = \frac{4}{(x+2)(x-2)}$?

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ELL English Language Learners (Use with EXAMPLE 3)

SPEAKING BEGINNING Tell students that this example is about identifying *extraneous solutions* to equations. Ask students to guess the meaning of *extraneous* and then use the word to describe things they might see around the classroom that could be viewed as *extraneous*.

Q: What smaller word do you see at the beginning of the word *extraneous*? [extra]

Q: What do you think *extraneous* means based on what you know about what *extra* means? [something extra that is not needed or useful]

LISTENING DEVELOPING Read the information in the call-out for Step 3 aloud to students. Have students listen carefully to the usage of the phrase *under the assumption*. If necessary, have students listen to other examples you provide using this phrase to help them understand its meaning more clearly.

Q: What is an *assumption*? [something that is accepted as true]

Q: What kind of information would you typically expect to get from a phrase using the word *under*? Is that how it's used here? [information about the position or location of an object; no]

WRITING EXPANDING Read the call-out for Step 4 with the students. Have students write in their journals the meaning of the term *valid*, as well as sentences using the term *valid*.

Q: When is a solution *valid*? [A solution is valid when it does not make the denominator of a fraction equal to 0.]

Q: What does *not valid* mean? [not reasonable or acceptable]



STEP 2 Understand & Apply

EXAMPLE 4 Check for Extraneous Solutions

Pose Purposeful Questions ETP

Q: How can you determine values that cannot be solutions to the equation without solving?

[Any value that makes a denominator zero cannot be a solution to the equation. You can set each denominator equal to zero to find these values.]

Q: Do you have to multiply both sides of the equation by the least common denominator to eliminate the fractions?

[You do not have to multiply by the least common denominator, but you do have to multiply by a common denominator.]

Have a Growth Mindset: Give Your Best Effort and Persist

When you give your best effort and persist, you have a better chance of reaching your goal. Ask: *What does it mean to persist? How do you know that you are getting better at something? How do you feel when you try hard and get better at something?*

Try It! Answers

4. a. $x = 2$
b. $x = 5$

Elicit and Use Evidence of Student Thinking ETP

Q: What do you notice about the equation in part (b)? How can this help you solve the problem?

[The denominators on each side of the equation are equal. Because the denominators are equal, you can set the numerators equal to each other to solve. Then check for extraneous solutions.]

Use with EXAMPLES 3 & 4

HABITS OF MIND

Communicate Precisely What is an extraneous solution?

[An extraneous solution is a solution that arises from manipulations of equivalent equations but is not a solution to the original problem.]

Examples are available at SavvasRealize.com.

EXAMPLE 4 Check for Extraneous Solutions

What are the solutions to the following equations?

A. $\frac{5x}{x-2} = 7 + \frac{10}{x-2}$

$$\frac{5x}{x-2} = 7 + \frac{10}{x-2}$$

$$(x-2)\left(\frac{5x}{x-2}\right) = \left(7 + \frac{10}{x-2}\right)(x-2)$$

$$5x = 7(x-2) + 10$$

$$5x = 7x - 14 + 10$$

$$-2x = -4$$

$$x = 2$$

Check the solution in the original equation. The value 2 is an extraneous solution because it would cause the denominator in the original equation to be equal to 0.

This equation has no solution.

B. $\frac{3}{x-3} = \frac{x}{x-3} - \frac{x}{4}$

$$\frac{3}{x-3} = \frac{x}{x-3} - \frac{x}{4}$$

$$(4)(x-3)\left(\frac{3}{x-3}\right) = \left(\frac{x}{x-3} - \frac{x}{4}\right)(4)(x-3)$$

$$4(3) = 4(x) - x(x-3)$$

$$12 = 4x - x^2 + 3x$$

$$x^2 - 7x + 12 = 0$$

$$(x-3)(x-4) = 0$$

$$x-3 = 0 \text{ or } x-4 = 0$$

$$x = 3 \text{ or } x = 4$$

Check the solutions in the original equation. The value 3 is an extraneous solution because it would cause the denominator of the original equation to be equal to zero. The only solution to the equation is $x = 4$.

Try It! 4. What are the solutions to the following equations?

a. $x + \frac{6}{x-3} = \frac{2x}{x-3}$

b. $\frac{x^2}{x+5} = \frac{25}{x+5}$

Support Struggling Students

USE WITH EXAMPLE 4 Students may need additional practice with multiplying rational expressions, when solving problems with extraneous solutions. Have students practice multiplying these rational expressions.

Find the product.

1. $\frac{x-4}{x+7} \cdot \frac{x+5}{x-4}$ $\left[\frac{x+5}{x+7}\right]$

2. $\frac{x^2-16}{x^2+6x+9} \cdot \frac{x^2+5x+6}{x^2-3x-4}$ $\left[\frac{(x+4)(x+2)}{(x+3)(x+1)}\right]$

3. $\frac{x^2+5x-14}{2x^2+7x-4} \cdot \frac{2x^2-9x+4}{x^2+9x+14}$ $\left[\frac{(x-2)(x-4)}{(x+2)(x+4)}\right]$



Activity

EXAMPLE 5 Solve a Rate Problem

Pose Purposeful Questions ETP

- Q:** What does each side of the equation represent?
[the amount of time Jake paddled in one direction]
- Q:** Why is the 1 added to $\frac{16}{5+c}$ rather than to $\frac{6}{5-c}$?
[$\frac{6}{5-c}$ represents Jake's time traveling upstream, and his time traveling upstream equals 1 hour more than his time traveling downstream.]
- Q:** Would $c = -25$ be an extraneous solution if you were solving the problem without context?
[No; $c = -25$ would not be an extraneous solution. This solution satisfies the original equation and does not cause the original equation to have a zero in the denominator.]

Try It! Answer

5. 75 minutes

Elicit and Use Evidence of Student Thinking ETP

- Q:** How would you set up an equation to solve this problem?
[You know that it takes 20 minutes for three people working together to plant the garden. You can add each person's rate and set it equal to the rate of the three people together.]

Common Error

Try It! 5 Some students may answer the question with the solution of the equation instead of using the solution to answer the question. Emphasize that $x = 60$ is the number of minutes it takes Benito to plant the garden on his own, not Tyler. Point out that Tyler takes $x + 15$, or 75 minutes. Remind students to make sure that they answered the question being asked.

Examples are available at SavvasRealize.com.

APPLICATION

EXAMPLE 5 Solve a Rate Problem

Paddling with the current in a river, Jake traveled 16 miles. Even though he paddled upstream for an hour longer than the amount of time he paddled downstream, Jake could only travel 6 miles against the current. In still water, Jake paddles at a rate of 5 mph. What is the speed of the current in the river?



Formulate Let c be the rate of the river's current in miles per hour. Recall that distance = rate $\times$ time, so time = $\frac{\text{distance}}{\text{rate}}$.

Jake's paddle rate in still water + the rate of the current: $\frac{16}{5+c} + 1 = \frac{6}{5-c}$
 Jake paddles 1 h longer upstream than downstream.

Compute Solve the equation for c .

$$\frac{16}{5+c} + 1 = \frac{6}{5-c}$$

$$(5+c)(5-c)\left(\frac{16}{5+c} + 1\right) = \left(\frac{6}{5-c}\right)(5+c)(5-c)$$

$$80 - 16c + 25 - c^2 = 30 + 6c$$

$$0 = c^2 + 22c - 75$$

$$0 = (c+25)(c-3)$$

$$0 = c+25 \quad 0 = c-3$$

$$c = -25 \quad c = 3$$

Interpret The solution $c = -25$ is extraneous because the speed of the current cannot be negative. The speed of the current is 3 mph.

Try It! 5. Three people are planting tomatoes in a community garden. Marta takes 50 minutes to plant the garden alone, Benito takes x minutes and Tyler takes $x + 15$ minutes. If the three of them take 20 minutes to finish the garden, how long would it have taken Tyler alone?

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Use with EXAMPLE 5

HABITS OF MIND

Make Sense and Persevere What does the fraction $\frac{1}{50}$ mean with regards to Marta?

[This fraction refers to the amount of garden Marta can plant in 1 minute.]

CONCEPT SUMMARY Solving Rational Equations

Q: How can you determine if a solution is in the domain?
[Substitute the solution into the original equation. The solution is in the domain when the denominators are nonzero and the equation is true.]

Do You UNDERSTAND? | Do You KNOW HOW?

Common Error

Exercise 6 Some students may say that the solutions to this equation are $x = 3$ and $x = -3$. Have students verify that each solution satisfies the original equation and does not cause the denominator to equal zero.

Answers

- Begin by finding the LCD of all the terms, then multiply each term by the LCD, then simplify. Lastly, substitute the solution into the original equation to see if any of the terms are undefined as a result. If so, that solution cannot be an element of the domain and is considered extraneous.
- Sample: $\frac{2}{x+1} - \frac{1}{x-1} = \frac{-2}{x^2-1}$
- The student did not ensure that the solution is valid. This solution is extraneous, and there is no solution to this equation.
- Once the original equation has been multiplied by the LCD, the solution might yield values that are not solutions to the original equation.
- $x = -4$
- $x = 3$
- 12 km/h



Concept Summary, Do You Understand, and Do You Know How are available at SavvasRealize.com.

CONCEPT SUMMARY Solving Rational Equations

WORDS A rational equation is an equation that contains a rational expression. To solve, identify the domain for the variable. Then multiply both sides of the equation by a common denominator and solve. An extraneous solution is a solution that is not valid because that value is excluded from the domain of the original equation.

ALGEBRA $\frac{1}{x} + \frac{2}{x} = \frac{1}{6}$ Domain: $x \neq 0$
 $6x\left(\frac{1}{x} + \frac{2}{x}\right) = 6x\left(\frac{1}{6}\right)$
 $6 + 12 = x$
 $18 = x$
 The domain includes $x = 18$, so the solution to the equation is 18.

$\frac{x^2+4}{x-1} = \frac{5}{x-1}$ Domain: $x \neq 1$
 $(x-1)\left(\frac{x^2+4}{x-1}\right) = (x-1)\left(\frac{5}{x-1}\right)$
 $x^2 + 4 = 5$
 $x^2 = 1$
 $x = \pm 1$
 The domain does not include $x = 1$, so 1 is an extraneous solution. It does include $x = -1$, so the solution to the equation is -1 .

Do You UNDERSTAND?

- ESSENTIAL QUESTION** How can you solve rational equations and identify extraneous solutions?
- Vocabulary** Write your own example of a rational equation that, when solved, has at least one extraneous solution.
- Error Analysis** A student solved the rational equation as follows:
 $\frac{1}{2x} - \frac{2}{5x} = \frac{1}{10x} - 3; x = 0$
 Describe and correct the error the student made.
- Construct Arguments** Yuki says, "You can check the solution(s) of rational equations in any of the steps of the solution process." Explain why her reasoning is incorrect.

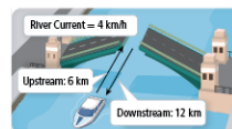
Do You KNOW HOW?

Solve.
 5. $\frac{4}{x+6} = 2$ 6. $\frac{x^2}{x+3} = \frac{9}{x+3}$

7. Organizing given information into a table can be helpful when solving rate problems. Use this table to solve the following problem.

	Distance	Rate	Time
Upstream			
Downstream			

A boat can travel 6 km upstream in the same time it takes to travel 12 km downstream. Find the speed of the boat in still water.





PRACTICE & PROBLEM SOLVING

Lesson Practice You may opt to have students complete the automatically scored Practice and Problem Solving items online powered by MathXL for School.

Choose from: Lesson Practice

Adaptive Practice

You may also take advantage of the bank of exercises for assigning additional practice.

Assignment Guide

On-Level	Advanced
8, 10, 11, 15–33	8–33

Engage Through Student Choice

Promote student agency by allowing students to choose practice items. You may structure this choice in many ways.

For example:

Assign each section a point value. Students choose at least one item from each section and items chosen should have a minimum of 20 total points.

Understand, Apply.....2 points each

Practice.....1 point each

Assessment Practice.....1 point each

Performance Task3 points each

Item Analysis

Example	Items	DOK
1	15–18, 32	1
	12	2
2	19, 20, 26	2
	8	3
3	21, 22	1
	11	2
4	23, 24, 31	1
	10, 14	2
	13	3
5	9, 25, 27–29	2
	30, 33	3



Practice & Problem Solving and video tutorials are available at SavvasRealize.com.

Scan the QR code to access a video tutorial.

PRACTICE & PROBLEM SOLVING

UNDERSTAND

8. Reason If you solve a work-rate problem and your solution, which represents the amount of time it would take *working together*, exceeds the individual *working alone* times that are given, then how do you know your solution is unreasonable? Explain.

9. Construct Arguments Explain why a negative solution must be eliminated as an extraneous solution when solving a rational equation for an unknown rate.

10. Error Analysis Describe and correct the error Miranda made in solving the rational equation.

$$\frac{1}{x-2} + \frac{x-2}{x+2} = \frac{x-4}{x-2}$$

$$(x+2)(x-2)\left(\frac{1}{x-2} + \frac{x-2}{x+2}\right) = (x+2)(x-2)\left(\frac{x-4}{x-2}\right)$$

$$(x+2)(1) + (x-2)(x-2) = (x+4)(x+2)$$

$$x+2+x^2-4x+4 = x^2+6x+8$$

$$-3x+6 = 6x+8$$

$$-2 = 9x; \text{ or } x = -\frac{2}{9}$$

11. Generalize In addition to identifying extraneous solutions, why else is it important to substitute your solution into the original equation?

12. Mathematical Connections Explain how solving rational equations is related to solving linear and quadratic equations.

13. Higher Order Thinking Write a rational equation that cannot have 2 or -6 as solutions.

14. Make Sense and Persevere Solve the rational equation shown. Explain what is unique about the solution.

$$\frac{x^2-7x-18}{x+2} = x-9$$

PRACTICE

Solve the equation. SEE EXAMPLE 1

$$15. \frac{1}{x-3} = 10$$

$$16. \frac{15}{x+3} = 3$$

$$17. \frac{12}{x-4} = 9$$

$$18. \frac{5}{3-x} = 1$$

Solve the problem. SEE EXAMPLE 2

19. Working individually, Paige and Malia can each complete a landscaping job in the time shown. Working together, how long would it take them to complete the job?



20. Russel and Aaron can build a shed in the time shown when working together. Aaron works three times as fast as Russel. How long would it take Russel to build the shed if he were to work alone?



Solve the equation. SEE EXAMPLE 3

$$21. \frac{x}{x-3} - 4 = \frac{3}{x-3}$$

$$22. \frac{x^2}{x-10} = \frac{100}{x-10} - 10$$

Solve the equation. SEE EXAMPLE 4

$$23. \frac{4}{3(x+1)} = \frac{12}{x^2-1}$$

$$24. \frac{x}{x-3} + \frac{2x}{x+3} = \frac{18}{(x+3)(x-3)}$$

Solve the problem. SEE EXAMPLE 5

25. A boat travels 8 miles upstream in the same amount of time it can travel 12 miles downstream. In still water the speed of the boat is 5 mi/h. What is the speed of the current?

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Answers

- Verify that students' answers are correct.
- Oftentimes the steps involved in solving a rational equation transform it into a linear or quadratic equation.
- Sample: $\frac{1}{x-2} + \frac{2}{x+6} = 10$
- Sample: The equation reduces to an identity; the solution is the set of all real numbers except -2.
- $x = \frac{31}{10}$

See answers for Exercises 16–25 on next page.



STEP 3 Practice & Problem Solving

Answers

16. $x = 2$
 17. $x = \frac{16}{3}$
 18. $x = -2$
 19. $2\frac{2}{5}$ hours
 20. 32 hours
 21. no solution
 22. $x = -20$
 23. $x = 10$
 24. $x = -2$
 25. The speed of the current is 1 mi/h.
 26. $1\frac{1}{5}$ h
 27. a.

	Distance	Rate	Time
With Wind	1500	$550 + r$	$\frac{1500}{550 + r}$
Against Wind	1000	$550 - r$	$\frac{1000}{550 - r}$

- b. $\frac{1500}{550 + r} = \frac{1000}{550 - r}$; $r = 110$ mi/h
 28. $n = 2$ friends
 29. 40 and 50 mi/h
 30. $h = 8$ hits
 33. Part A 150 gallons

Part B Using the graphing function of your calculator, enter $2.5 + 0.05x$ for y_1 , and $1 + 0.06x$ for y_2 . Then use the TABLE function to find the y -values of the function that are equivalent when $x = 150$.



Practice & Problem Solving is available at SavvasRealize.com.

PRACTICE & PROBLEM SOLVING

APPLY

26. **Make Sense and Persevere** Kenji can finish a puzzle in 2 hours working alone. Oscar can finish the same puzzle in 3 hours working alone. How long would it take Oscar and Kenji to finish the puzzle if they worked on it together?
 27. **Use Structure** A commercial jet flies 1,500 miles with the wind. In the same amount of time it can fly 1,000 miles against the wind. The speed of the jet in still air is 550 mph. Find the speed of the wind.



- a. Organize the given information and what you need to find in a table.
 b. Write and solve a rational equation to find the wind speed.
 28. **Reason** During their day at the beach, Jae and his friends rent a Jet Ski. They split the \$120 rental fee evenly among themselves. Then Jae, with only his friend Orion, share the cost of a \$16 pizza. If Jae spends a total of \$48 for both, then find the number of friends, n , with whom he shared the cost of the Jet Ski rental.
 29. **Make Sense and Persevere** When driving to their family reunion, Rio's mom drove 10 miles at a rate of x mph and then 25 miles at a rate of $x + 10$ mph. The total driving time was 45 minutes. What were the two driving speeds at which Rio's mom drove?
 30. **Generalize** So far this baseball season, Philip has gotten a hit 8 times out of 40 at-bats. He wants to increase his batting average to 0.333. Calculate the number of consecutive hits, h , he would need in order to achieve this goal. Round your answer to the nearest whole number.

ASSESSMENT PRACTICE

31. Which of the following rational equations have at least one extraneous solution? Select all that apply.
☐ $\frac{2}{x} = \frac{3}{x-4}$
☒ $\frac{x^2}{x-3} = \frac{9}{x-3}$
☐ $\frac{x-1}{x-5} = \frac{9}{x-5}$
☐ $x + \frac{3}{x} = 4$
☒ $\frac{x}{x-3} - \frac{3}{2} = \frac{3}{x-3}$
 32. **SAT/ACT** Which of the following is the solution of $\frac{3}{x+1} = \frac{2}{x-3}$?
☒ A $x = -11$
☐ B $x = -\frac{7}{5}$
☐ C $x = \frac{7}{5}$
☐ D $x = 11$
 33. **Performance Task** A chemist needs alcohol solution in the correct concentration for her experiment. She adds a 6% alcohol solution to 50 gallons of solution that is 2% alcohol. The function that represents the percent of alcohol in the resulting solution is $f(x) = \frac{50(0.02) + x(0.06)}{50 + x}$, where x is the amount of 6% solution added.



Part A How much 6% solution should be added to create a solution that is 5% alcohol?

Part B **Use Appropriate Tools** Explain the steps you could take to use your graphing calculator to verify the correctness of your answer to part (A).

LESSON 4-5 Solving Rational Equations 219



Assessment

LESSON QUIZ

Use the Lesson Quiz to assess students' understanding of the mathematics in the lesson.

Students can take the Lesson Quiz online or you can download a printable copy from [SavvasRealize.com](https://www.savvasrealize.com). The Lesson Quiz is also available in the *Assessment Sourcebook*.

Item Analysis

Item	DOK	Skills Review and Practice
1	2	A87
2	2	A87
3	2	A87
4	2	A87
5	2	A87



Use the student scores on the Lesson Quiz to prescribe differentiated assignments.

If students take the Lesson Quiz online, it will be automatically scored and appropriate differentiated practice will be assigned based on student performance.

I Intervention	0–3 points	• Reteach to Build Understanding • Mathematical Literacy and Vocabulary • Lesson Virtual Nerd videos
O On-Level	4 points	• Enrichment
A Advanced	5 points	• Enrichment



Lesson Quiz is available at [SavvasRealize.com](https://www.savvasrealize.com).

Assessment Resources

Name _____

enVision Algebra 2

SavvasRealize.com

4-5 Lesson Quiz

Solving Rational Equations

1. Solve the equation $\frac{1}{2x-3} = 4$.

☐ A $\frac{1}{2}$
☐ B $\frac{8}{13}$
☐ C $\frac{11}{8}$
☒ D $\frac{13}{8}$

2. Aaron can paint a door in 6 hours, and Anna can do the same job in 3 hours. Working together, how many hours will it take them to paint a door?

$h =$

3. Solve the equation $\frac{3}{x-1} = \frac{4}{2x+3}$.

$x =$

4. Solve the equation $\frac{-3}{x+3} = \frac{x}{x+3} - \frac{x}{5}$. Identify any extraneous solutions.

solution: $x =$

extraneous solution: $x =$

5. In still water, the speed of a boat is 10 mi/h. Camilla drives the boat 5 miles upstream and 5 miles back to her starting point in an hour and 20 minutes. What is the speed of the stream?

☐ A 2 mi/h
☐ B 3 mi/h
☐ C 4 mi/h
☒ D 5 mi/h

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